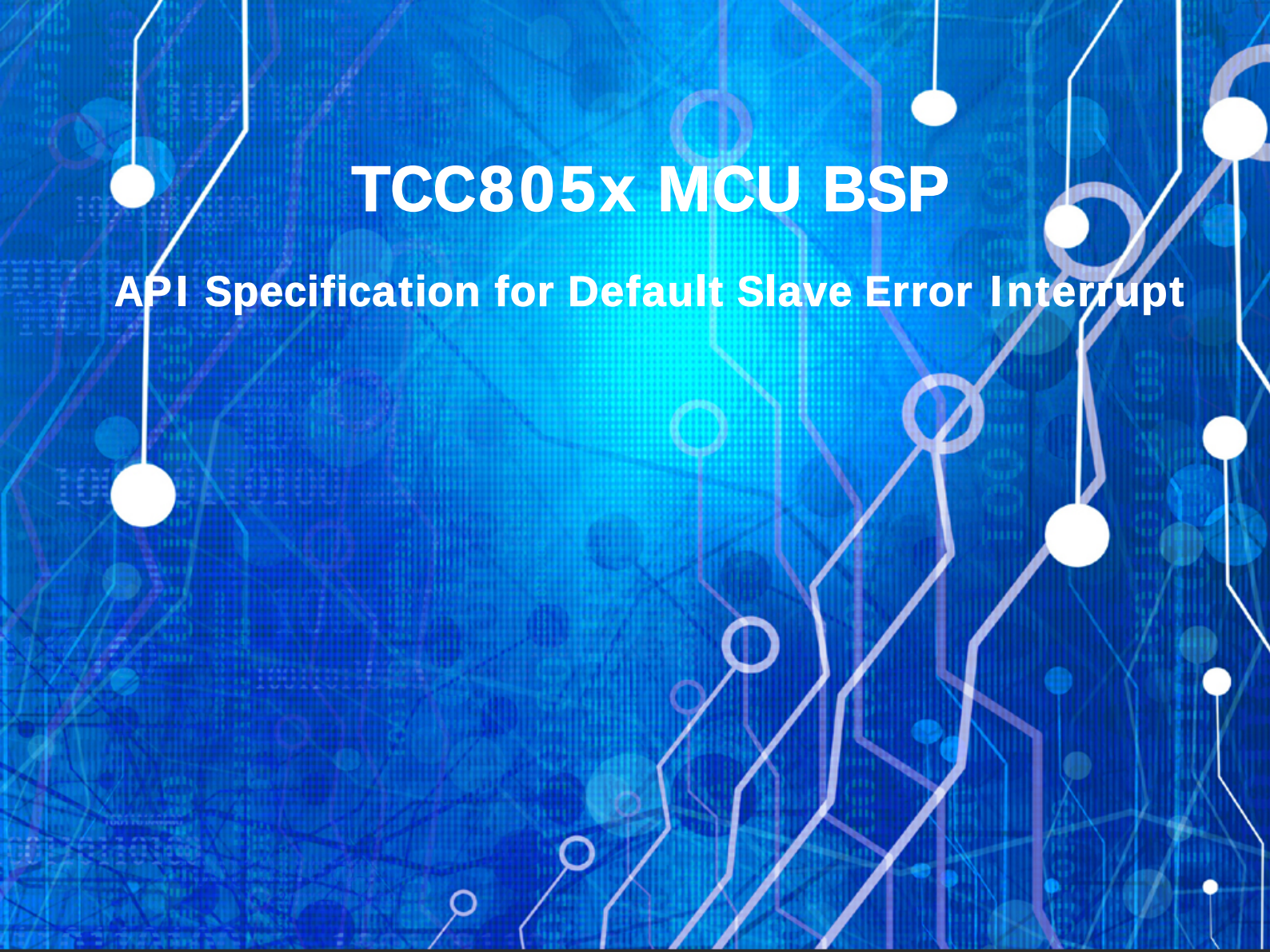




TCC805x MCU BSP

API Specification for Default Slave Error Interrupt

Rev. 1.00 [A]

2020-11-12

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1 INTRODUCTION

This document describes the default slave error driver (hereinafter referred to as DSE driver) of TCC805x MCU BSP. For more detailed information of the default slave error interrupt handler in the TCC805x MCU Sub-system, refer to Chapter 25 Default Slave Error Interrupt Handler of the *“PART 11-MICOM SUB-SYSTEM”* in the *“TCC805x Full Specification”*. [1]

2 TERMS AND ABBREVIATIONS

Table 2.1 Terms and Abbreviation

DSE	Default Slave Error
------------	---------------------

3 SOURCE FILES

3.1 Location of Source Files

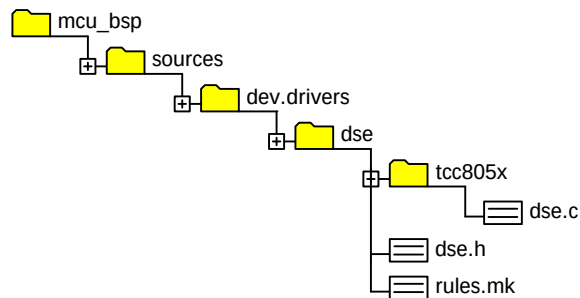


Figure 3.1 Location of Source Files

3.2 Simple Description of Source Files

- `dse.h`
 - API header of DSE driver
- `dse.c`
 - Driver source of DSE driver
- `rules.mk`
 - Make rules

4 CONSTANTS

4.1 Default Slave Error Configuration Flag Values

These constants are used to determine which DSE module will be enabled.

Declaration

```
#define DSE_INTERCON      0x00000001u
#define DSE_AP_MBOX       0x00000002u
#define DSE_SYSPERI       0x00000004u
#define DSE_UART          0x00000008u
#define DSE_PDM           0x00000010u
#define DSE_MC_MBOX       0x00000020u
#define DSE_CAN           0x00000040u
#define DSE_GPSB          0x00000080u
#define DSE_ISOLATE       0x00000100u
#define DES_SEL_ALL       (DSE_INTERCON | DSE_AP_MBOX | DSE_SYSPERI | DSE_UART | DSE_PDM |
                           DSE_MC_MBOX | DSE_CAN | DSE_GPSB | DSE_ISOLATE)
#define DES_SEL_MASK      DES_SEL_ALL
#define DSE_INT_MODE      0x00000000u    /* Default Slave Error interrupt mode */
#define DSE_RES_MODE      0x10000000u    /* Default Slave Error response mode */
#define DSE_RES_INT_MODE  0x20000000u    /* Default Slave Error int & res mode */
#define DSE_MODE_MASK     0x30000000u
```

Description

Value	Description
DSE_INTERCON	Enable Peripheral AHB Interconnector's default slave error handler.
DSE_AP_MBOX	Enable AP Mailbox Terminal AHB Multiplexer's default slave error handler.
DSE_SYSPERI	Enable System Peripheral AHB Multiplexer's default slave error handler.
DSE_UART	Enable UART AHB Interconnector's default slave error handler.
DSE_PDM	Enable PDM AHB Multiplexer's default slave error handler.
DSE_MC_MBOX	Enable MCU Mailbox Terminal AHB Multiplexer's default slave error handler.
DSE_CAN	Enable CAN AHB Interconnector's default slave error handler.
DSE_GPSB	Enable GPSB AHB Interconnector's default slave error handler.
DSE_ISOLATE	Enable Incoming AHB Isolator's default slave error handler.
DES_SEL_ALL	Enable all of the default slave error handler.
DES_SEL_MASK	Mask value for selecting module of the default slave error handler in the <i>DSE_Init</i> input parameter.

Value	Description
<i>DSE_INT_MODE</i>	<i>Enable Error IRQ of the default slave error handler.</i>
<i>DSE_RES_MODE</i>	<i>Enable Error Response of the default slave error handler. DSE_MC_MBOX and DSE_MC_MBOX do <u>not</u> support response mode.</i>
<i>DSE_RES_INT_MODE</i>	<i>Enable Error IRQ and Error Response of the default slave error handler.</i>
<i>DSE_MODE_MASK</i>	<i>Mask value for selecting mode of the default slave error handler in the DSE_Init input parameter.</i>

Remark

In response mode (*DSE_RES_MODE* & *DSE_RES_INT_MODE*), when the CPU receives an error response, data abort occurs in the CPU. In IRQ mode (*DSE_INT_MODE* & *DSE_RES_INT_MODE*), the CPU receives the IRQ from the default slave error IRQ handler.

4.2 Register Duplication Check Configuration Values

These constants are used for checking register duplication.

Declaration

```
#define DSE_SM_IRQ_MASK      0x2u
#define DSE_SM_IRQ_EN       0x4u
#define DSE_SM_SLV_CFG      0x8u
#define DSE_SM_CFG_WR_PW    0x10000u
#define DSE_SM_CFG_WR_LOCK  0x20000u
#define DSE_SM_SF_CHK_GRP_SEL (DSE_SM_IRQ_MASK | DSE_SM_IRQ_EN | DSE_SM_SLV_CFG
                               | DSE_SM_CFG_WR_PW | DSE_SM_CFG_WR_LOCK)
```

Description

Value	Description
<i>DSE_SM_IRQ_MASK</i>	<i>Enable Peripheral AHB Interconnector's default slave error handler.</i>
<i>DSE_SM_IRQ_EN</i>	<i>Enable AP Mailbox Terminal AHB Multiplexer's default slave error handler.</i>
<i>DSE_SM_SLV_CFG</i>	<i>Enable System Peripheral AHB Multiplexer's default slave error handler.</i>
<i>DSE_SM_CFG_WR_PW</i>	<i>Enable UART AHB Interconnector's default slave error handler.</i>
<i>DSE_SM_CFG_WR_LOCK</i>	<i>Enable PDM AHB Multiplexer's default slave error handler.</i>
<i>DSE_SM_SF_CHK_GRP_SEL</i>	<i>Enable MCU Mailbox Terminal AHB Multiplexer's default slave error handler.</i>

Remark

None

5 STRUCTURES

5.1 DSEReg_t

The DSE register is handled by using the following structure.

Declaration

```
typedef struct _DSEReg_
{
    uint32 rDSE_IRQ_STATUS;
    uint32 rDSE_IRQ_MASK;
    uint32 rDSE_IRQ_EN;
    uint32 rDEF_SLV_CFG;
    uint32 rDEF_SLV0_ADDR;
    uint32 rDEF_SLV1_ADDR;
    uint32 rDEF_SLV2_ADDR;
    uint32 rDEF_SLV3_ADDR;
    uint32 rDEF_SLV4_ADDR;
    uint32 rDEF_SLV5_ADDR;
    uint32 rDEF_SLV6_ADDR;
    uint32 rDEF_SLV7_ADDR;
    uint32 rDEF_SLV8_ADDR;
    uint32 rDEF_RESERVED[3];
    uint32 rDSE_IH_CFG_WR_PW;
    uint32 rDSE_IH_CFG_WR_LOCK;
    uint32 rSF_CHK_GRP_EN;
    uint32 rSF_CHK_GRP_STS;
} DSEReg_t;
```

Description

Value	Description
<i>rDSE_IRQ_STATUS</i>	Default Slave Error Interrupt Status.
<i>rDSE_IRQ_MASK</i>	Default Slave Error Interrupt Mask.
<i>rDSE_IRQ_EN</i>	Default Slave Error Interrupt Enable.
<i>rDEF_SLV_CFG</i>	Default Slave Interrupt/Error Response Configuration.
<i>rDEF_SLV0_ADDR</i>	When the <i>rDSE_IRQ_STATUS</i> is <i>DSE_INTERCON</i> , <i>rDEF_SLV0_ADDR</i> indicates the traffic address that caused the error.
<i>rDEF_SLV1_ADDR</i>	When the <i>rDSE_IRQ_STATUS</i> is <i>DSE_AP_MBOX</i> , <i>rDEF_SLV1_ADDR</i> indicates the traffic address that caused the error.
<i>rDEF_SLV2_ADDR</i>	When the <i>rDSE_IRQ_STATUS</i> is <i>DSE_AP_SYSPERI</i> , <i>rDEF_SLV2_ADDR</i> indicates the traffic address that caused the error.
<i>rDEF_SLV3_ADDR</i>	When the <i>rDSE_IRQ_STATUS</i> is <i>DSE_UART</i> , <i>rDEF_SLV3_ADDR</i> indicates the traffic address that caused the error.

Value	Description
<i>rDEF_SLV4_ADDR</i>	When the <i>rDSE_IRQ_STATUS</i> is <i>DSE_PDM</i> , <i>rDEF_SLV4_ADDR</i> indicates the traffic address that caused the error.
<i>rDEF_SLV5_ADDR</i>	When the <i>rDSE_IRQ_STATUS</i> is <i>DSE_MC_MBOX</i> , <i>rDEF_SLV5_ADDR</i> indicates the traffic address that caused the error.
<i>rDEF_SLV6_ADDR</i>	When the <i>rDSE_IRQ_STATUS</i> is <i>DSE_CAN</i> , <i>rDEF_SLV6_ADDR</i> indicates the traffic address that caused the error.
<i>rDEF_SLV7_ADDR</i>	When the <i>rDSE_IRQ_STATUS</i> is <i>DSE_GPSB</i> , <i>rDEF_SLV7_ADDR</i> indicates the traffic address that caused the error.
<i>rDEF_SLV8_ADDR</i>	When the <i>rDSE_IRQ_STATUS</i> is <i>DSE_ISOLATE</i> , <i>rDEF_SLV8_ADDR</i> indicates the traffic address that caused the error.
<i>rDEF_RESERVED</i>	Not used.
<i>rDSE_IH_CFG_WR_PW</i>	If you want to write to <i>rDSE_IH_CFG_WR_LOCK</i> or <i>rSF_CHK_GRP_EN</i> , you should first write a predefined password value (0x5AFEACE5) to <i>rDSE_IH_CFG_WR_PW</i> . If you successfully write the predefined value(0x5AFEACE5) to <i>rDSE_IH_CFG_WR_PW</i> , <i>rDSE_IH_CFG_WR_PW</i> is read as 1.
<i>rDSE_IH_CFG_WR_LOCK</i>	Default Slave Error Interrupt Handler Configuration Write Lock. When <i>rDSE_IH_CFG_WR_LOCK</i> is written as 1, writing of the following values is prohibited. ■ <i>rDSE_IRQ_MASK</i> ■ <i>rDSE_IRQ_EN</i> ■ <i>rDEF_SLV_CFG</i>
<i>rSF_CHK_GRP_EN</i>	Default Slave Error Interrupt Handler Configuration Soft Fault Check Group Enable. Refer to Chapter 4.2 Register Duplication Check Configuration Values.
<i>rSF_CHK_GRP_STS</i>	Default Slave Error Interrupt Handler Configuration Soft Fault Check Group Status. If an error occurs in the register set in <i>rSF_CHK_GRP_EN</i> , you can read this value to identify which register caused the error.

Remark

None

6 APIs

6.1 DSE_Init

Initializes the Default Slave Error Interrupt Handler block.

Declaration

```
SALRetCode_t DSE_Init  
(  
    uint32 uiModeFlag  
);
```

Parameters

Value	Description
uiModeFlag	[in] Refer to Chapter 4.1.

Return

Value	Description
SAL_RET_SUCCESS	Success
SAL_RET_FAILED	Failure

Remark

None

6.2 DSE_Deinit

De-initializes the Default Slave Error Interrupt Handler block.

Declaration

```
void DSE_Deinit  
(  
    void  
);
```

Parameters

None

Return

None

Remark

None

6.3 DSE_WriteLock

Enables or disables the default slave error register write protection.

Declaration

```
SALRetCode_t DSE_WriteLock  
(  
    uint32 uiLock  
);
```

Parameters

Value	Description
uiLock	[in] Write Protection 1: Enable, 0: Disable.

Return

Value	Description
SAL_RET_SUCCESS	Success
SAL_RET_FAILED	Failure

Remark

None

6.4 DSE_GetGrpSts

Get the *SF_CHK_GRP_STS* register value. Refer to [DSEReg_t](#).

Declaration

```
uint32 DSE_GetGrpSts
(
    void
);
```

Parameters

None

Return

Value	Description
<i>DSE_SM_IRQ_MASK</i>	Unintended changes occur in the <i>DSE_IRQ_MASK</i> register.
<i>DSE_SM_IRQ_EN</i>	Unintended changes occur in the <i>DSE_IRQ_EN</i> register.
<i>DSE_SM_SLV_CFG</i>	Unintended changes occur in the <i>DEF_SLV_CFG</i> register.
<i>DSE_SM_CFG_WR_PW</i>	Unintended changes occur in the <i>DSE_IH_CFG_WR_PW</i> register.
<i>DSE_SM_CFG_WR_LOCK</i>	Unintended changes occur in the <i>DSE_IH_CFG_WR_LOCK</i> register.

Remark

None

7 HOW TO USE DSE DRIVER

You can use the DSE driver just by calling the *DSE_Init(DES_SEL_ALL)* function. The current code is designed to output a debug message when a default slave error interrupt occurs.

7.1 Example of Using DSE Register Duplicate Check

The following example shows how to enable DSE's register duplicate check. For detailed description of register setting, refer to Chapter 26 System Safety Mechanism (SM) Control Module of the “*PART 11-MICOM SUB-SYSTEM*” in the “*TCC805x Full Specification*”. [1]

```
void DSE_EnableRegisterDuplicateCheck(void)
{
    #if defined(FMU_DRIVER)
        uint32 uiRdata    = 0UL;
        uint32 uiErr      = 0UL;
        uint32 uiI        = 0UL;

        DSE_Init(DES_SEL_ALL);
        (void)FMU_IsrHandler((FMUFaultid_t)FMU_ID_SYS_SM_CFG, (FMUSeverityLevelType_t)FMU_SVL_LOW,
            (FMU_INT_FNCT_PTR)&DSE_TEST_Fmu_IRQ_ISR, NULL_PTR);
        (void)FMU_Set((FMUFaultid_t)FMU_ID_SYS_SM_CFG);
        (void)FMU_IsrClr((FMUFaultid_t)FMU_ID_SYS_SM_CFG);

        SAL_WriteReg(0x5AFEACE5UL, DSETEST_SM_SF_CTRL_PW);    // 0x1B9211C0UL
        uiRdata = SAL_ReadReg(DSETEST_SM_SF_CTRL_PW);

        if(uiRdata != 1UL)
        {
            printf(" Fail.. DSETEST_SM_SF_CTRL_PW set\n");
            uiErr |= (0x10UL);
        }
        else
        {
            uiRdata = SAL_ReadReg(DSETEST_SM_SF_CTRL_CFG);    // 0x1B9211F0UL
            SAL_WriteReg(uiRdata | 0x88UL, DSETEST_SM_SF_CTRL_CFG);    // 0x1B9211F0UL
            printf(" SF_CTRL_CFG set\n");
        }
    }

    static void DSE_TEST_Fmu_IRQ_ISR(void *pArg)
    {
        uint32 uiRdata    = 0UL;
        uint32 uiReadSts   = 0UL;

        (void)pArg;
        uiReadSts = SAL_ReadReg(DSETEST_SM_SF_CTRL_STS); //0x1B9211F4UL
        printf("FMU interrupt ... SF_CTRL_STS x%x\n", uiReadSts);

        if(uiReadSts == 0x8UL)
        {
            uiRdata = DSE_GetGrpSts();
        }
    }
}
```

```
    if((uiRdata & DSE_SM_IRQ_MASK) == DSE_SM_IRQ_MASK)
    {
        printf("DSE_IRQ_MASK error\n");
    }
    if((uiRdata & DSE_SM_IRQ_EN) == DSE_SM_IRQ_EN)
    {
        printf("DSE_SM_IRQ_EN error\n");
    }
    if((uiRdata & DSE_SM_SLV_CFG) == DSE_SM_SLV_CFG)
    {
        printf("DSE_SM_SLV_CFG error\n");
    }
    if((uiRdata & DSE_SM_CFG_WR_PW) == DSE_SM_CFG_WR_PW)
    {
        printf("DSE_SM_CFG_WR_PW error\n");
    }
    if((uiRdata & DSE_SM_CFG_WR_LOCK) == DSE_SM_CFG_WR_LOCK)
    {
        printf("DSE_SM_CFG_WR_LOCK error\n");
    }
}
}
```

8 REFERENCES

- [1] TCC805x Full Specification
- [2] Contact Telechips for more details: sales@telechips.com

9 REVISION HISTORY

Rev. 1.00: 2020-11-12

- First version release

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